Statistical models

Mock exam

Hamburg University of Technology

Prof. Dr. Nihat Ay and Dr. Matthias Lienau

Summer Term 2025

July 3, 2025

The idea of this mock exam is to show you how the exam could look like. Please be aware that there will be topics in the final exam that were not covered in the mock exam and vice versa.

Problem 1 (6 Points)
Decide if the following statements are true or false. Each correct answer gives one point, while there is a deduction of 0.5 points for every false answer. The total number of points from this problem cannot become negative.

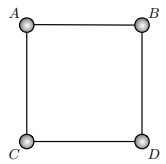
1) Let X and Y be independent random variables such that $X \sim \chi^2_m$ with $m \in \mathbb{N}$ and $Y \sim \text{Normal}(0,1)$. Then, $X/Y \sim t_m$.
☐ true ☐ false

2) Let $(\varphi_n)_{n\in\mathbb{N}}$ be a sequence of tests such that under the null hypothesis the probability of a type I error converges to $\alpha \in (0,1)$. Then, $(\varphi_n)_{n\in\mathbb{N}}$ are asymptotic tests with level α.
☐ true ☐ false

3) In a Gaussian linear model with intercept and $n > k+1$, where n is the sample size and k is the number of independent variables, the residuals $\widehat{\varepsilon}$ satisfy $\|\widehat{\varepsilon}\|^2/\sigma^2 \sim \chi^2_{n-k-1}$.
☐ true ☐ false

4) Let X_1, X_2 be two variables that model two dice rolls (dice with six possible outcomes). The statistical model consisting of distributions for which the two variables are independent has dimension 36.
☐ true ☐ false

5) The following figure illustrates four variables A, B, C, D with distribution that factorises according to the shown graph.



The conditional independence statement $A \perp\!\!\!\perp D | B$ holds true.
☐ true ☐ false

6) Consider a finite set I, a statistical model $\mathcal{M}$ in $\mathcal{P}_+(I)$, and a probability measure μ in $\mathcal{P}_+(I)$. Assume that there exists a μ^* in $\mathcal{M}$ that minimises the KL-divergence $D_{\text{KL}}(\mu\|\nu)$ with respect to $\nu \in \mathcal{M}$. Then μ^* is unique.
☐ true ☐ false

Problem 2 (10 Points)
The ages of the players of three football teams A, B und C are given. In order to study if there are differences between the ages of the players in the teams, an analysis of variance is considered.

In order to answer the questions below, you have to use the following output in `R`:

```
Call:
lm(formula = age ~ team, data = Football)

Residuals:
    Min       1Q  Median      3Q     Max
-8.2220  -2.2996  0.4645  2.1030  6.4011

Coefficients:
             Estimate  Std. Error  t value  Pr(>|t|)
(Intercept)   26.9965      0.6895   39.154   < 2e-16  ***
teamB          2.5845      0.9751    2.650   0.00988  **
teamC         -0.9060      0.9751   -0.929   0.35590
- - -
Signif.  codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’  0.1 ‘ ’ 1

Residual standard error:  3.447 on 72 degrees of freedom
Multiple R-squared:  0.1609, Adjusted R-squared:  0.1375
F-statistic:  6.901 on 2 and 72 DF, p-value:  0.001812
```

a) State the assumptions on the ages of the players of the teams A, B and C for analysis of variance.

b) Provide the sample means of the ages of the players of the teams A, B and C as well as the estimate for the standard deviation.

c) Check if there is a difference between ages of players of different teams. Formulate the null hypothesis and the alternative and use a test with level 0.05.

Moreover, the following output in `R` is given:

```
Tukey multiple comparisons of means
95% family-wise confidence level

Fit:  aov(formula = lm(age ~ team, Football))

$team
           diff         lwr        upr      p adj
B-A   2.5844612   0.2509559   4.917967  0.0263971
C-A  -0.9060279  -3.2395332   1.427477  0.6237408
C-B  -3.4904891  -5.8239945  -1.156984  0.0017773
```

d) Decide which teams are different with respect to the age of the players. Formulate the test problems and use the Tukey-Kramer test with level 0.05.

Problem 3 (10 Points)

a) In a physics experiment one investigates the radioactivity R at time t, which is given by

$$R = \alpha_0 \exp(-\alpha_1 t) + \varepsilon$$

with $\alpha_0, \alpha_1 \in \mathbb{R}$ and a random error ε such that $\mathbf{E}[\varepsilon] = 0$ and $\mathbf{E}[\varepsilon^2] < \infty$. It can be assumed that the random errors for different measurements are uncorrelated. The following data were observed:

t	0	1	2	3
R	10	7.5	5	3

Formulate the optimisation problem that is solved by a least squares estimator $(\widehat{\alpha}_0, \widehat{\alpha}_1)^T$ for $(\alpha_0, \alpha_1)^T$ explicitly with the given numbers.

b) It is studied if some mechanical part is destroyed by a given force or not. Let I be one if the part is destroyed and zero otherwise and let f denote the force. Logistic regression is used to explain I by an intercept β_0 and the force f with coefficient β_1. Provide the probability that the part is destroyed in terms of β_0, β_1 and f.

c) Consider the Gaussian linear model

$$Y = \gamma_0 + \gamma_1 x^{(1)} + \gamma_2 x^{(2)} + \varepsilon$$

with $\varepsilon \sim \text{Normal}(0, \sigma^2)$. Assume that one has n observations, that the observations of Y are given by $\mathbf{Y}$ and that the design matrix $\mathbf{A}$ satisfies

$$\mathbf{A}^T\mathbf{A} = \begin{pmatrix} n & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{pmatrix}.$$

Let $\widehat{\gamma}_0, \widehat{\gamma}_1, \widehat{\gamma}_2$ and $\widehat{\sigma}$ be the usual estimators for $\gamma_0, \gamma_1, \gamma_2$ and σ and let $\alpha \in (0,1)$. Show that

$$\varphi(\mathbf{Y}) = \begin{cases} \text{cannot reject}, & 0 \in (\widehat{\gamma}_1 - \widehat{\gamma}_2 - t_{n-3,1-\alpha/2}\widehat{\sigma}, \widehat{\gamma}_1 - \widehat{\gamma}_2 + t_{n-3,1-\alpha/2}\widehat{\sigma}), \\ \text{reject}, & 0 \notin (\widehat{\gamma}_1 - \widehat{\gamma}_2 - t_{n-3,1-\alpha/2}\widehat{\sigma}, \widehat{\gamma}_1 - \widehat{\gamma}_2 + t_{n-3,1-\alpha/2}\widehat{\sigma}), \end{cases}$$

is a level-α test for

$$H_0 : \gamma_1 = \gamma_2 \quad \text{against} \quad H_1 : \gamma_1 \neq \gamma_2.$$

Problem 4 (10 Points)
Let $I = \{1, 2, 3\}$. Consider $\mu \in \mathcal{M}_+(I)$ given by

$$\mu \;=\; 1\,\delta^1 + 2\,\delta^2 + 3\,\delta^3$$

and $a, b \in \mathcal{S}(I)$ (the tangent space of $\mathcal{M}_+(I)$ for all base points $\mu \in \mathcal{M}_+(I)$) given by

$$a \;=\; 2\,\delta^1 + 4\,\delta^2 + 2\,\delta^3, \qquad b \;=\; -5\,\delta^1 + 2\,\delta^2 + 1\,\delta^3.$$

Compute with respect to the Fisher-Rao inner product in μ:

1. the inner product $\langle a, b \rangle_\mu$,
2. the norms $\|a\|_\mu$, $\|b\|_\mu$,
3. the cosine similarity of a and b,
4. the angle between a and b in degrees.

Problem 5 (10 Points)
Let $I = \{1, 2, 3\}$. Consider $\mu_1, \mu_2 \in \mathcal{P}_+(I)$ given by

$$\mu_1 \;=\; \frac{1}{6}\,\delta^1 + \frac{1}{3}\,\delta^2 + \frac{1}{2}\,\delta^3, \qquad \mu_2 \;=\; \frac{1}{2}\,\delta^1 + \frac{1}{3}\,\delta^2 + \frac{1}{6}\,\delta^3.$$

Compute for μ_1, μ_2:

1. the Hellinger distance,
2. the Bhattacharya coefficient,
3. the Fisher-Rao distance,
4. the Kullback-Leibler divergence $D_{\mathrm{KL}}(\mu_1 \| \mu_2)$.